

Discipline	Category	Subcategory	Element	Attributes	Spec	Unit	Cost (USD)
Civil	Site Works	Grading	Cut / Fill	Cut (excavate material, stockpile on site)	Bulk excavation for pads/trenches	yd³	\$12 – \$25 / yd³
Civil	N/A	N/A		Fill (import or re-use, place & compact)	Placing, compacting to spec (compaction testing)	yd³	\$18 – \$35 / yd³
Civil	N/A	N/A	Finished grade	Fine grading (final tolerance, proof roll)	Fine grading to elevations & tolerances	ft²	\$0.10 – \$0.75 / ft²
Civil	N/A	N/A	Slopes	Slope shaping, benches, cut/fill slope development	2:1 to 3:1 typical slopes; benched or keyed	LF (slope length)	\$6 – \$18 / LF
Civil	N/A	N/A	Erosion control	Temporary erosion controls (silt fence, inlet protection)	Silt fence, inlet filters, EPS / rock checks	LF or LS	\$3 – \$8 / LF (silt fence)
Civil	N/A	N/A	Topsoil	Topsoil strip (salvage) & re-spread or supply & place imported topsoil	4–6 in topsoil layer	yd³	\$8 – \$18 / yd³
Civil	N/A	Utilities	Site lighting	Pole-mounted or wall-mounted fixtures	Includes conduit & circuiting	EA	\$1,200 – \$5,000 / pole
Civil	N/A	Containment	Secondary containment	110% tank containment (liner + curb)	Coated concrete curb or HDPE-lined basin	CY (contain vol) / EA	\$80 – \$250 / ft³ (liner & basin) OR \$8,000 – \$25,000 / EA depending on approach
Civil	N/A	Drainage	Grassed swale construction	Shape & seed swale, geotextile if required		LF or SY	\$6 – \$25 / LF
Civil	Site Works	Demolition	Remove small structure / pavement	Break out & haul off		SY or LS	\$5 – \$25 / SY or \$1,000 – \$8,000 LS
Civil	Access	Roads / Paving	Site roads	Pavement type selection (asphalt 4" over 8" agg base)	12 ft lane typical; includes subgrade prep	SY	\$12 – \$30 / SY
Civil	N/A	N/A	N/A	Aggregate base (8")	Crushed stone, compacted	SY	\$6 – \$12 / SY
Civil	N/A	N/A	N/A	Curb & gutter or edge restraint	Concrete curb, edge detailing	LF	\$25 – \$70 / LF
Civil	N/A	N/A	Drainage	Roadside/inlet drains & connection to swale	Cast-inlet or grate inlet	EA	\$800 – \$3,000 / inlet
Civil	Access	Pavement markings	Striping & signage	Thermoplastic striping & signs		LS	\$800 – \$3,000
Civil	Stormwater	Retention	Basins / Channels	Detention basin excavation	On-site shallow basin, liner if required	yd³	\$15 – \$35 / yd³
Civil	N/A	N/A	N/A	Detention structure (concrete or modular tank)	Underground tank or concrete vault	ft³ or EA	\$150 – \$450 / ft³ (vault)
Civil	N/A	N/A	N/A	Outlet controls (orifice / orifice plate / riser)	Outlet control structure with manhole	EA	\$1,500 – \$8,000 / EA
Civil	N/A	N/A	N/A	SWPPP implementation	SWPPP plan, inspection & O&M items	LS	\$2,500 – \$10,000 / LS
Civil	N/A	Pipework	Storm drain piping	HDPE / PVC / RCP storm piping	6"–12" small conveyance	LF	\$25 – \$120 / LF
Civil	Utilities - Site	Duct banks	Electrical ductbank	Conduit size (e.g., 4" or 3×2" conduit)	4" PVC/HDPE conduits typical; 24" min cover	LF	\$60 – \$150 / LF (installed)
Civil	N/A	N/A	N/A	Pull chambers / access vaults	Precast or cast-in-place manholes every 100–200 ft	EA	\$2,000 – \$8,000 / EA
Civil	N/A	N/A	N/A	Concrete encasement / warning tape	Concrete surround & tape	LF	\$10 – \$40 / LF
Civil	Hydrology	Culverts / Drainage	Culverts	Culvert pipe (RCP / HDPE) — size	12"–24" typical for small crossings	LF	\$60 – \$250 / LF (installed, size dependent)
Civil	N/A	N/A	N/A	Invert & headwall construction	Headwall, end treatments, riprap scour protection	EA	\$1,000 – \$6,000 / EA
Civil	N/A	N/A	N/A	Scour protection / riprap	Riprap placement at inlets/outlets	CY	\$75 – \$250 / CY

Civil	Survey	Topography	Benchmarking	Datum & coordinate system establishment	Set 3–6 control points, tie to local datum	EA / LS	\$200 – \$800 / control pt (or \$3,000–\$8,000 LS for full topo & control)
Civil	N/A	N/A	Site topographic survey	Topo with contours and utilities	0.1–0.5 acre small site topo	LS	\$2,500 – \$8,000 / LS
Civil	N/A	N/A	As-built survey	Post-construction as-built	As-built of foundations, piping, pavement extents	LS	\$1,500 – \$5,000 / LS
Civil	Geotechnical	Soils & Foundations	Borings & lab tests	2 borings + lab testing & report (prelim)	Standard penetration, lab gradation, Atterberg	LS	\$4,000 – \$9,000 / LS